💡 Conductivity Lab: What Lets Electricity Through?: Lab Sheet

Chemistry · Unit 1C · Level 3 · goes with L1.13 No Swimming, L1.13.5 You Light Up My Life

Name: _____

Question: How does dissolving a substance in water affect whether it conducts electricity?

Materials

conductivity tester (low voltage), foil or copper, wax, sugar, salt, distilled water, cups, balance and graduated cylinder

⚠️ Safety

- Goggles on the whole lab.
- Never taste anything, even salt or sugar.
- Only the low-voltage tester. Nothing in wall outlets.
- Wash hands after.

Prediction

Predict which materials will conduct. Explain using electrons or ions.

_____

_____

Procedure

1. Check that the tester bulb lights when its two ends touch.
2. Test each solid: foil/copper, wax, sugar, salt crystals. Record.
3. Make 100 mL salt water at 1%, 5% and 10% (1 g, 5 g, 10 g of salt). Test each.
4. Test distilled water and sugar water.
5. Dry the tester between every test.
6. Sort the materials into conductors and insulators.

Data

Material	Prediction	Bulb: bright / dim / off	Conductor or insulator?
foil / copper			
wax			
salt crystals			
distilled water			
1% salt water			
10% salt water			
sugar water			

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does dissolving salt in water affect conductivity?

2. Evidence: Use your data for distilled water, 1% and 10% salt water.

3. Reasoning: Why does sugar water NOT conduct? Use the word ions.

4. **Current to charges: 1 A = 6.28×10^{18} charges per second.**

> Example: 0.2 A = $0.2 \times 6.28 \times 10^{18} = 1.256 \times 10^{18}$ charges per second.

a) 0.5 A = ____ × 10^{18} charges/s _____

b) Which moves more charge each second: 0.1 A or 0.4 A? _____